Ananya Gupta

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EDUCATION

Bachelor of Science in Electrical Engineering *Expected Fall 2028*
Rutgers University–New Brunswick, New Brunswick, NJ | GPA: 3.947
Relevant Coursework: Circuit Theory, Radio Frequency, Electromagnetism, Signal Processing, Materials Science, Digital Systems, Verilog, Quantum Computing, Computer Science

TECHNICAL SKILLS

Engineering Software: SolidWorks, Ansys (Fluent CFD, Static Structural, Mechanical), MATLAB, KiCad, XFLR5, Excel
Test Equipment: Oscilloscopes, Multimeters, Spectrum Analyzers
RF & Hardware: Circuit Design & Prototyping, PCB Design, Impedance Matching & Resonance Tuning, Soldering, Motor Drivers & Sensors, Arduino, ESP32, Raspberry Pi, LoRa, ESP-NOW, CAN Bus, UART
Materials & Manufacturing: Composite Fabrication, Epoxy Handling, 3D Printing, Solar Panel Assembly
Programming & Tools: Python, C++, Git, Linux, OpenCV, MediaPipe, WebSockets

EXPERIENCE

RF Design Engineer | IEEE *August 2025 – May 2026*
- Designed and prototyped an RF telegraph using circuit theory, transmitter/receiver modules, and Arduinos: Morse code sent over a 433 MHz link and displayed on an LCD
- Ran iterative RF tests with teammates to refine device performance, turning engineering theory into manufacturable hardware

Design & Manufacturing Lead | AIAA Aviation *August 2025 – May 2026*
- Led design and manufacturing of a V-tail competition aircraft (foam wings, carbon tail boom, 3D-printed fuselage) through a full redesign and repeated iterations, reaching 110 mph; tracked requirements and followed build and test procedures throughout
- Fabricated lightweight composite structural components and built production timelines to keep fabrication on schedule
- Wrote proposals and presented preliminary design reviews to judges, using SolidWorks, XFLR5 stability simulations, and MATLAB to justify aerodynamic and motor/propeller choices

Solar Array & Telemetry Team Member | Rutgers Solar Car Club *August 2026 – Present*
- Run Ansys Fluent CFD simulations to study how adding solar panels affects a car's aerodynamics, iterating the geometry to improve it; solder and assemble solar panels to support vehicle design and energy strategy decisions
- Built Wi-Fi telemetry: an ESP32 reads the car's CAN bus data over UART and streams it over WebSockets to a browser dashboard for real-time monitoring and debugging; evaluating LoRa for long-range transmission

Robotics Team Member | RU HART *August 2026 – Present*
- Building an autonomous delivery robot with a team: wrote PWM motor driver code, calibrated motors, and integrated sensors
- Building a fail-safe wireless E-stop remote with two ESP32-S3 boards over ESP-NOW (custom PCB planned); developing autonomous capabilities with OpenCV and a web server interface

Engineering Team Lead | SWEMatch *August 2025 – May 2026*
- Led a team of 6 through the full product lifecycle (requirements, system design, testing, prototyping) of a low-cost assistive technology device
- Architected NeuroLens in Python: real-time emotion recognition running on-device on a Raspberry Pi Zero 2W, using a compact CNN quantized to ~178 KB and WebSockets for live results

PROJECTS

Guitar Preamplifier PCB | KiCad
- Designing a guitar preamplifier PCB in KiCad, from schematic capture through board layout

AI-Powered 3D Modeling Tool | Python, FastAPI, build123d, Three.js
- Built a web app that turns plain-English prompts into 3D CAD parts, using tested involute spur-gear math and a sandboxed runner with timeouts; exports print-ready STL and STEP files

PC Parts Price Comparison Website | JavaScript, HTML/CSS
- Built a web app comparing landed prices for 14 PC components across 6 retailers, ranked by performance per dollar